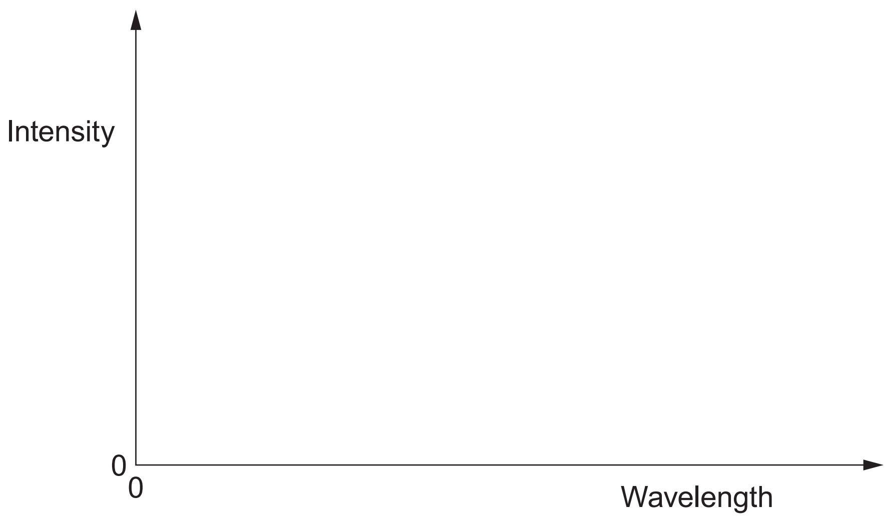
Examiner only

Option B – **Medical Physics**

8. (a) **Sketch two graphs** to show how the intensity of X-rays from an X-ray tube varies with wavelength for an X-ray tube operating at two different voltages, one at 30 kV and the other at 50 kV.

Label the main features of the graphs and also show which graph is at the higher voltage.
Space for calculations. [4]



(b) (i) Ultrasound can be used to measure the speed of blood through an artery. If ultrasound of frequency 1.5 MHz travelling at an angle of 30° to the direction of blood flow shows a Doppler shift of 200 Hz, calculate the speed of blood flow. The speed of ultrasound through the blood is 1570 m s^{-1}. [2]

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(ii) Suggest why it is important for doctors to monitor the rate of blood flow in a patient. [1]

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(c) (i) A magnetic resonance scanner (MRI) can be used to detect tumours in a patient's body. Describe how an MRI scanner works. [3]

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(ii) The MRI scanner has a magnetic field of 1.4 T. Determine the wavelength of electromagnetic radiation that should be used to detect the tumour. [2]

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(d) (i) PET scanners are often used to diagnose tumours in patients. Calculate the energy of the emitted gamma rays ($m_{\text{positron}} = m_{\text{electron}} = 0.000\,549\,\text{u}$, $1\,\text{u} = 931\,\text{MeV}$). [2]

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(ii) Explain why PET scanners are not commonly found in smaller hospitals. [1]

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Examiner only

(e) Doctors are concerned about the growth of an unborn baby and have the choice of the following imaging techniques.

X-ray **fluoroscopy** **ultrasound B-scan** **ultrasound A-scan** **CT scan**

Evaluate the suitability of **all five** types of imaging techniques for unborn babies. [5]

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